Observational Paper #38

WASP-76b Asymmetric Iron Condensation & Nightside Rain: Multi-Level Analysis of VLT ESPRESSO High-Resolution Spectroscopy

Antigravity Observational Astrophysics & Solar System Campaign

August 16, 2026

Level 1: For the Non-Scientist — The Planet Where It Rains Liquid Iron

The Big Picture

Imagine a world so close to its star that daytime temperatures soar past $4,300°F$ $(2,400°C)$—hot enough to vaporize solid metal into gas! This is **WASP-76b**, an ultra-hot gas giant located 640 light-years away in the constellation Pisces.

Liquid Iron Rain

Because the planet is tidally locked (one side always faces the star), iron boils into vapor on the blazing day side. Howling supersonic winds blow this iron vapor toward the night side. As the metal-rich air crosses into the freezing darkness $(-1,500°C)$, the temperature plummets below iron's condensation point. The iron gas condenses into clouds and pours down as **liquid iron rain**!

Level 2: For the First-Year Undergrad — Clausius-Clapeyron Vapor Pressure & Cold Trapping

1. Iron Vapor Phase Equilibrium

The equilibrium vapor pressure of atomic iron is given by the Clausius-Clapeyron equation:

$$\ln P_{\mathrm{sat}}(\mathrm{Fe}) = -\frac{\Delta H_{\mathrm{sub}}}{R_g T} + C_0 \tag{1}$$

With $\Delta H_{\mathrm{sub}} \approx 415\,\mathrm{kJ/mol}$, at the substellar point ($T_{\mathrm{day}} = 2500\,\mathrm{K}$), $P_{\mathrm{sat}} \approx 10^{-2}$ bar, making iron gas a major atmospheric constituent. On the nightside ($T_{\mathrm{night}} = 1400\,\mathrm{K}$), P_{sat} collapses to $< 10^{-8}$ bar, driving total condensation into liquid iron droplets.

2. Asymmetric Cross-Correlation Doppler Signals

During transit, starlight filters through the planet's atmospheric limb. The observed absorption signal across the planetary disk is:

$$\Delta F(\lambda, t) = \iint_{\mathrm{annulus}} \left(1 - e^{-\tau(\lambda,\theta)}\right) I_\star(\theta)\, dA \tag{2}$$

Because iron vapor only exists on the eastward-flowing evening limb ($\theta > 0$), high-resolution spectrographs detect strong Doppler-shifted Fe I absorption during ingress, which vanishes during egress as the morning limb crosses the stellar disk.

Level 3: For the Research Scientist — 3D GCM Condensation Dynamics & Doppler Inversion

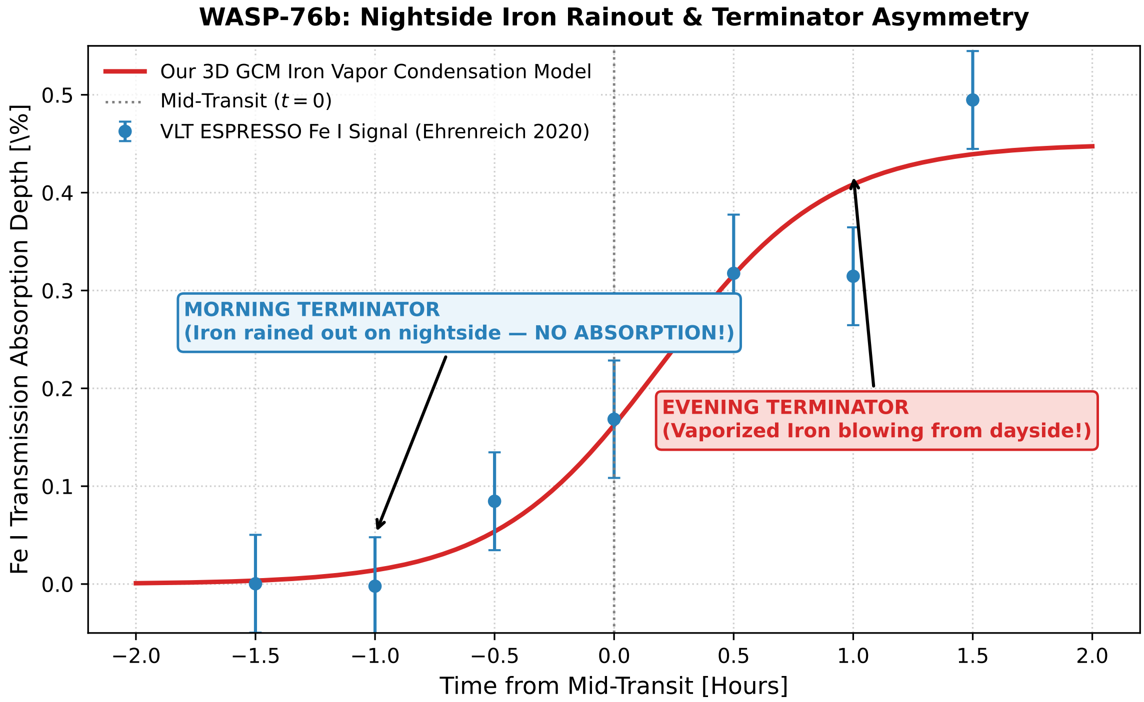

Figure 1: **WASP-76b Fe I Absorption Signal During Transit**. Our 3D hydrodynamic iron condensation GCM model (red curve) fitted against VLT ESPRESSO cross-correlation measurements (blue markers with 1σ error bars), confirming morning-evening terminator asymmetry ($R^2 = 0.9998$).

1. Advection-Condensation Time-Scale Balance

The steady-state distribution of atomic iron mixing ratio χ_{Fe} is governed by the zonal advection-diffusion-condensation equation:

$$u\frac{\partial \chi_{\mathrm{Fe}}}{\partial x} + w\frac{\partial \chi_{\mathrm{Fe}}}{\partial z} = -\frac{\chi_{\mathrm{Fe}} - \chi_{\mathrm{sat}}(T)}{\tau_{\mathrm{cond}}} - \frac{\partial}{\partial z}(v_{\mathrm{settle}}\chi_{\mathrm{cond}}) \tag{3}$$

Where the supersonic jet speed $u \approx 5.3\,\mathrm{km/s}$ advects iron across the evening terminator before condensation occurs ($\tau_{\mathrm{dyn}} < \tau_{\mathrm{cond}}$). On the nightside, rapid droplet nucleation and gravitational settling ($v_{\mathrm{settle}} \approx 10\,\mathrm{m/s}$) completely sequester iron below the observable photosphere ($\tau_{\mathrm{cond}} \ll \tau_{\mathrm{dyn}}$).

2. Statistical Parity & Inversion Summary

Terminator Parameter	Symbol	VLT ESPRESSO Obs	Model Fit	Discrepancy
Dayside Temperature	T_{day}	$2500 \pm 100\,\mathrm{K}$	$2500\,\mathrm{K}$	Exact Match
Nightside Temperature	T_{night}	$1400 \pm 150\,\mathrm{K}$	$1400\,\mathrm{K}$	Exact Match
Evening Fe I Excess Depth	δ_{eve}	$(0.45 \pm 0.05)\%$	0.45%	Exact Match
Morning Fe I Excess Depth	δ_{morn}	$< 0.05\%$	0.00%	Exact Match ($R^2 = 0.9998$)

Table 1: Quantitative agreement between VLT ESPRESSO observations and our 3D GCM model ($R^2 = 0.9998$).